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Inventor(s)	Nakanishi; Masaharu et al.

Semiconductor device

Abstract

A semiconductor device includes: a switching element including a drain electrode, a gate electrode, and a source electrode; a base supporting the switching element; and a first terminal, a second terminal, a third terminal, and a fourth terminal that each extend in the same direction. The switching element includes a temperature detection diode having a first electrode provided on the element obverse surface. Each of the drain electrode, the gate electrode, and the source electrode is electrically connected to a corresponding one of the first terminal, the second terminal, and the third terminal. The first electrode is electrically connected to the fourth terminal via a first wire.

Inventors:	**Nakanishi; Masaharu (Kyoto, JP), Kawamoto; Noriaki (Kyoto, JP)**
Applicant:	**ROHM CO., LTD.** (Kyoto, JP)
Family ID:	**1000008818777**
Assignee:	**ROHM CO., LTD. (Kyoto, JP)**
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Primary Examiner: Menz; Laura M

Attorney, Agent or Firm: HSML P.C.

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a semiconductor device.

BACKGROUND ART

(2) Patent document 1 discloses an example of a switching device (semiconductor device) equipped with a switching element such as a MOSFET. The switching element includes a drain electrode, a gate electrode, and a source electrode. The semiconductor device disclosed in the document includes three terminals (i.e., a gate terminal, a source terminal, and a drain terminal). The gate terminal is for inputting an electrical signal to the gate electrode. While the current to be converted based on the electrical signal is applied to the drain electrode via the drain terminal, a current converted based on the electrical signal will flow from the source electrode to the outside via the source terminal.

(3) In a semiconductor device such as the one disclosed in the patent document 1, it is necessary to accurately know the temperature of a bonding portion (i.e., junction temperature) of a semiconductor element because the presence/absence of a failure, life, and reliability are closely related to the temperature during operation. When the semiconductor element is a MOSFET, the junction temperature can be measured by using a body diode in the switching element and a thermal resistance measuring device. For example, when the thermal resistance measuring device is used, the junction temperature can be estimated by first applying a driving voltage to the switching element, then supplying a current to the body diode, and measuring the voltage with the thermal resistance measuring device.

(4) However, conducting the measurement of the junction temperature with the thermal resistance measuring device as described above is suitable in a laboratory, but not in a situation where the semiconductor device is actually used (i.e., when the switching element is driven).

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: JP-A-2019-121745

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) The present disclosure has been conceived in view of the problem noted above, and aims to provide a semiconductor device capable of measuring the junction temperature when a switching element is driven.

(7) A semiconductor device provided by the present disclosure includes: a switching element having an element obverse surface and an element reverse surface that face away from each other in a first direction, and including a drain electrode, a gate electrode, and a source electrode, where the switching element performs on/off control between the drain electrode and the source electrode by applying a driving voltage across the gate electrode and the source electrode with a potential difference being present between the drain electrode and the source electrode; a base having an obverse surface and a reverse surface that face away from each other in the first direction, and supporting the switching element with the element reverse surface facing the obverse surface; and a first terminal, a second terminal, a third terminal, and a fourth terminal that each extend in a second direction perpendicular to the first direction. The switching element includes a temperature detection diode having a first electrode provided on the element obverse surface. Each of the drain electrode, the gate electrode, and the source electrode is electrically connected to a different one of the first terminal, the second terminal, and the third terminal. The first electrode is electrically connected to the fourth terminal via a first wire.

Advantages of the Invention

(8) According to the configuration as described above, the junction temperature can be measured when the switching element is driven.

(9) Other features and advantages of the present disclosure will be more apparent from the detailed description given below with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view showing a semiconductor device according to a first embodiment.
(2) FIG. **2** is a plan view showing a semiconductor device A**1** in FIG. **1**.
(3) FIG. **3** is a cross-sectional view taken along line III-III in FIG. **2**.
(4) FIG. **4** is a cross-sectional view taken along line IV-IV in FIG. **2**.
(5) FIG. **5** is a cross-sectional view taken along line V-V in FIG. **2**.
(6) FIG. **6** shows the circuit configuration of the semiconductor device according to the first embodiment.
(7) FIG. **7** is a plan view showing a variation of the semiconductor device according to the first embodiment.
(8) FIG. **8** is a plan view showing a variation of the semiconductor device according to the first embodiment.
(9) FIG. **9** is a plan view showing a semiconductor device according to a second embodiment.
(10) FIG. **10** is a cross-sectional view taken along line X-X in FIG. **9**.
(11) FIG. **11** is a cross-sectional view taken along line XI-XI in FIG. **9**.
(12) FIG. **12** is a plan view showing a semiconductor device according to a third embodiment.
(13) FIG. **13** is a cross-sectional view taken along line XIII-XIII in FIG. **12**.
(14) FIG. **14** is a plan view showing a semiconductor device according to a fourth embodiment.
(15) FIG. **15** is a cross-sectional view taken along line XV-XV in FIG. **14**.
(16) FIG. **16** is a plan view showing a semiconductor device according to a fifth embodiment.
(17) FIG. **17** is a plan view showing a variation of the semiconductor device in FIG. **12**.
(18) FIG. **18** is a cross-sectional view taken along line XVIII-XVIII in FIG. **17**.

MODE FOR CARRYING OUT THE INVENTION

(19) The following describes preferred embodiments of the present disclosure with reference to the drawings.

(20) FIGS. **1** to **5** show a semiconductor device according to a first embodiment of the present disclosure. A semiconductor device A**1** according to the present embodiment includes a switching element **1**, a lead frame **2**, a gate wire **52**, a source wire **53**, a first wire **61**, a second wire **62**, and a sealing resin **7**.

(21) FIG. **1** is a perspective view showing the semiconductor device A**1**. FIG. **2** is a plan view showing the semiconductor device A**1**. FIG. **3** is a cross-sectional view taken along line III-III in FIG. **2**. FIG. **4** is a cross-sectional view taken along line IV-IV in FIG. **2**. FIG. **5** is a cross-sectional view taken along line V-V in FIG. **2**. In FIG. **2**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line. For convenience of understanding, the thickness direction of the semiconductor device A**1** is defined as a first direction z, the vertical direction in a plan view (FIG. **2**), which is perpendicular to the first direction z, is defined as a second direction y, and the horizontal direction in a plan view (FIG. **2**), which is perpendicular to both of the first direction z and the second direction y, is defined as a third direction x. Note that the terms “upper” and “lower” in the following description are used for convenience, and do not limit the posture of the semiconductor device A**1** during installation.

(22) The switching element **1** is made with the use of Si or Sic as a base material, and realizes the switching function of the semiconductor device A**1**. Examples of the switching element **1** include a SiC-MOSFET (Metal-Oxide-Semiconductor Field Effect Transistor), a SiC-bipolar transistor, a SiC-JFET (Junction Field Effect Transistor), and a SiC-IGBT (Insulated Gate Bipolar Transistor).

In the present embodiment, description is given of the case where the switching element **1** is a SiC-MOSFET.
(23) As shown in FIGS. **3** and **4**, an element obverse surface **11** is the upper surface of the switching element **1**. An element reverse surface **12** is the lower surface of the switching element **1**. The element obverse surface **11** and the element reverse surface **12** face away from each other in the first direction z.
(24) As shown in FIGS. **2** to **5**, the switching element **1** includes a drain electrode **131**, a gate electrode **132**, and a source electrode **133**. The switching element **1** includes a temperature sensor. In the illustrated example, the temperature sensor is a temperature detection diode **15**, but the present disclosure is not limited to this.
(25) The drain electrode **131** is arranged on the element reverse surface **12**. The gate electrode **132** is arranged on the element obverse surface **11** (the surface opposite to the surface on which the drain electrode **131** is arranged). The source electrode **133** is arranged on the element obverse surface **11** (the same surface as the surface on which the gate electrode **132** is formed). The source electrode **133** is larger than the gate electrode **132**. The switching element **1** applies a driving voltage to the gate electrode **132** and the source electrode **133** with a potential difference between the drain electrode **131** and the source electrode **133**, thereby performing the on/off control of the drain electrode **131** and the source electrode **133**.
(26) The temperature detection diode **15** includes a pn junction diode portion **150** built into the switching element **1** by a semiconductor process, first electrode **151**, and a second electrode **152**. In the present embodiment, the pn junction diode portion **150** is formed near the element obverse surface **11**, and the first electrode **151** and the second electrode **152** are arranged on the element obverse surface **11**. In the present embodiment, the first electrode **151** is an anode electrode, and the second electrode **152** is a cathode electrode.
(27) In the present embodiment, the switching element **1** has a rectangular shape as viewed in the thickness direction (as viewed in the first direction z). For example, the switching element **1** has a dimension of 1 mm to 10 mm square as viewed in the first direction z. The switching element **1** has a dimension of 40 μm to 700 μm in the thickness direction, for example.
(28) The switching element **1** is supported by a die pad **20** described below via a bonding member **3**. The bonding member **3** is a conductive bonding member formed with the use of TiNiAg solder, SnAgCu solder, Pb solder, or calcined Ag, for example, so as to electrically connect the drain electrode **131** of the switching element **1** and the die pad **20**.
(29) The lead frame **2** is a conductive member and bonded to a circuit board (not illustrated) to form a conductive path between the switching element **1** and the circuit board. The lead frame **2** is made of an alloy mainly containing Cu. A part of the surface may be plated for corrosion resistance, electrical conductivity, thermal conductivity, or a bonding property, for example. The lead frame **2** includes the die pad **20**, a first terminal **21**, a second terminal **22**, a third terminal **23**, and a fourth terminal **24**, which are all made of the same lead frame material.
(30) The die pad **20** has an obverse surface **20***a* and a reverse surface **20***b*. The obverse surface **20***a* is the upper surface of the die pad **20**. The obverse surface **20***a* is the surface on which the switching element **1** is mounted. As shown in FIGS. **3** to **5**, the element reverse surface **12** of the switching element **1** faces the obverse surface **20***a*. The reverse surface **20***b* is the lower surface of the die pad **20**. The obverse surface **20***a* and the reverse surface **20***b* are flat and face away from each other in the first direction z.
(31) In the present embodiment, the die pad **20** is formed with a through hole **20***c* extending from the obverse surface **20***a* to the reverse surface **20***b*. The through hole **20***c* is spaced apart from the switching element **1** as viewed in the thickness direction (as viewed in the first direction z). In the present embodiment, the through hole **20***c* has a circular shape as viewed in the thickness direction, but the shape thereof is not particularly limited. The die pad **20** is an example of a “base”.
(32) The first terminal **21**, the second terminal **22**, the third terminal **23**, and the fourth terminal **24**

are spaced apart from each other in the third direction x, and are used when the semiconductor device A**1** is mounted on, e.g., a circuit board (not illustrated).
(33) As shown in FIG. **2**, the first terminal **21** is spaced apart from the die pad **20**, and extends along the second direction y. The first terminal **21** is arranged at the outermost position (left side in the figure) in the third direction x as viewed in the first direction z. The first terminal **21** has a first pad **211** and a tip portion **212**. The first pad **211** is closest to the die pad **20** in the second direction y. The tip portion **212** is the tip of the first terminal **21**, which is located on the opposite side from the first pad **211**, and is farthest from the die pad **20** in the second direction y. The gate wire **52** is bonded to the first pad **211**. The first terminal **21** is electrically connected to the gate electrode **132** via the gate wire **52**. In the present embodiment, the first terminal **21** is the gate terminal of the semiconductor device A**1**.
(34) The second terminal **22** is spaced apart from the die pad **20**, and extends along the second direction y. The second terminal **22** is arranged at the outermost position (right side in the figure) in the third direction x as viewed in the first direction z. As a result, the second terminal **22** and the first terminal **21** are located at the outermost positions that are opposite to each other in the third direction x. The second terminal **22** has a second pad **221** and a tip portion **222**. The second pad **221** is closest to the die pad **20** in the second direction y. The tip portion **222** is the tip of the second terminal **22**, which is located on the opposite side from the second pad **221**, and is farthest from the die pad **20** in the second direction y. The source wire **53** is bonded to the second pad **221**. The second terminal **22** is electrically connected to the source electrode **133** via the source wire **53**. In the present embodiment, the second terminal **22** is the source terminal of the semiconductor device A**1**. The second wire **62** is bonded to the second pad **221**. The second terminal **22** is electrically connected to the second electrode **152** via the second wire **62**.
(35) The third terminal **23** is connected to the die pad **20**, and extends along the second direction y from the die pad **20**. In the present embodiment, as shown in FIG. **2**, the third terminal **23** is connected to one end of the die pad **20** in the second direction y (lower end in the figure) at the center of the die pad **20** in the third direction x, as viewed in the first direction z. The third terminal **23** is positioned between the first terminal **21** and the second terminal **22** in the third direction x. The third terminal **23** has an intermediate bent portion **233** and a tip portion **232**. As shown in FIG. **5**, the intermediate bent portion **233** is a portion of the third terminal **23** that is bent such that a portion of the third terminal **23** exposed from the sealing resin **7** is shifted upward in the figure from the die pad **20** in the first direction z. The tip portion **232** is the tip of the third terminal **23**, and is farthest from the die pad **20** in the second direction y. The third terminal **23** is electrically connected to the drain electrode **131** via the die pad **20** and the bonding member **3**. In the present embodiment, the third terminal **23** is the drain terminal of the semiconductor device A**1**.
(36) The fourth terminal **24** is spaced apart from the die pad **20**, and extends along the second direction y. The fourth terminal **24** is positioned between the second terminal **22** and the third terminal **23** as viewed in the first direction z. The fourth terminal **24** has a fourth pad **241**, a tip portion **242**, and a bent portion **243**. The fourth pad **241** is closest to the die pad **20** in the second direction y. The tip portion **242** is the tip of the fourth terminal **24**, which is located on the opposite side from the fourth pad **241**, and is farthest from the die pad **20** in the second direction y. The bent portion **243** is positioned between the fourth pad **241** and the tip portion **242**, and is closer to the fourth pad **241** in the second direction y.
(37) As can be understood from FIGS. **1** and **2**, the tip side of the fourth terminal **24** beyond the bent portion **243** is shifted to one side in the first direction z (the side to which the obverse surface **20***a* of the die pad **20** faces). With the fourth terminal **24** having the bent portion **243**, the tip portion **242** of the fourth terminal **24** is shifted to the one side in the first direction z (the side to which the obverse surface **20***a* of the die pad **20** faces) as compared to the tip portions **212**, **222**, and **232** of the first terminal **21**, the second terminal **22**, and the third terminal **23**. An imaginary line in FIG. **1** indicates the shape of the fourth terminal **24** different from the shape of the fourth

terminal **24** in the present embodiment. Specifically, the imaginary line in FIG. **1** indicates the shape of the fourth terminal **24** when the fourth terminal **24** does not have the bent portion **243** and extends straight along the second direction y from the fourth pad **241** to the tip portion **242**.

(38) The first wire **61** is bonded to the fourth pad **241** of the fourth terminal **24**. The fourth terminal **24** is electrically connected to the first electrode **151** via the first wire **61**.

(39) As shown in FIG. **2**, the distance (first distance d**13**) in the third direction x between the center line C**1** of the first terminal **21** (gate terminal) and the center line C**3** of the third terminal **23** (drain terminal) is larger than the distance (second distance d**34**) in the third direction x between the center line C**3** of the third terminal **23** (drain terminal) and the center line C**4** of the fourth terminal **24**. The first distance d**13** is also larger than the distance (third distance d**24**) in the third direction x between the center line C**4** of the fourth terminal **24** and the center line C**2** of the second terminal **22** (source terminal). In the present embodiment, the second distance d**34** and the third distance d**24** are substantially the same. Also, the sum of the second distance d**34** and the third distance d**24** is substantially the same as the first distance d**13**.

(40) As shown in FIG. **2**, the gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the first pad **211** of the first terminal **21**, and electrically connects the gate electrode **132** of the switching element **1** and the first terminal **21** to each other. In FIG. **4**, the gate wire **52** is omitted.

(41) The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the second pad **221** of the second terminal **22**, and electrically connects the source electrode **133** of the switching element **1** and the second terminal **22** to each other. In FIGS. **4** and **5**, the source wire **53** is omitted.

(42) The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the fourth pad **241** of the fourth terminal **24**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **24** to each other. In FIG. **5**, the first wire **61** is omitted.

(43) The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the second pad **221** of the second terminal **22**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the second terminal **22** to each other.

(44) The gate wire **52**, the source wire **53**, the first wire **61**, and the second wire **62** are made of aluminum (Al), an Al alloy, Cu, or a Cu alloy, for example. Note that the source wire **53** may have a larger diameter than the other wires **52**, **61**, and **62**. It is possible to provide a plurality of source wires **53**.

(45) The sealing resin **7** covers and protects the switching element **1**, a part of the lead frame **2**, the gate wire **52**, the source wire **53**, the first wire **61**, and the second wire **62**. Specifically, of the lead frame **2**, the sealing resin **7** covers the die pad **20**, a part of the first terminal **21** (mainly the first pad **211**), a part of the second terminal **22** (mainly the second pad **221**), a part of the third terminal **23** (mainly the intermediate bent portion **233**), and a part of the fourth terminal **24** (mainly the fourth pad **241**). The sealing resin **7** is a thermosetting synthetic resin that is electrically insulative. The material of the sealing resin **7** is not particularly limited. For example, the sealing resin **7** may be made of a black epoxy resin and mixed with fillers as appropriate.

(46) In the present embodiment, the sealing resin **7** has a resin obverse surface **71**, a resin reverse surface **72**, a pair of resin first side surfaces **73**, and a pair of resin second side surfaces **74**. The resin obverse surface **71** is the upper surface of the sealing resin **7** shown in FIGS. **3** to **5**, and faces in the same direction as the obverse surface **20***a* of the die pad **20**. The resin reverse surface **72** is the lower surface of the sealing resin **7** shown in FIGS. **3** to **5**, and faces in the same direction as the reverse surface **20***b* of the die pad **20**. The resin obverse surface **71** and the resin reverse surface **72** face away from each other in the first direction z.

(47) As shown in FIG. **5**, the pair of resin first side surfaces **73** are spaced apart from each other in

the second direction y. The pair of resin first side surfaces **73** face away from each other in the second direction y. As shown in FIG. **5**, the upper end of each resin first side surface **73** is connected to the resin obverse surface **71**, and the lower end of each resin first side surface **73** is connected to the resin reverse surface **72**. In the present embodiment, the first terminal **21**, the second terminal **22**, the third terminal **23**, and the fourth terminal **24** are partially exposed from one of the resin first side surfaces **73**.

(48) As shown in FIGS. **3** and **4**, the pair of resin second side surfaces **74** are spaced apart from each other in the third direction x. The pair of resin second side surfaces **74** face away from each other in the third direction x. As shown in FIGS. **3** and **4**, the upper end of each resin second side surface **74** is connected to the resin obverse surface **71**, and the lower end of each resin second side surface **74** is connected to the resin reverse surface **72**.

(49) As shown in FIG. **1**, the sealing resin **7** is formed with a pair of recesses **75** that are recessed into the sealing resin **7** from the upper portions of the pair of resin second side surfaces **74**. As shown in FIGS. **1** and **5**, the sealing resin **7** is formed with a resin through hole **76** extending from the resin obverse surface **71** to the resin reverse surface **72**. In the present embodiment, the center of the resin through hole **76** coincides with the center of the through hole **20***c* in the die pad **20**. The diameter of the resin through hole **76** is smaller than the diameter of the through hole **20***c*. In the present embodiment, the entire wall of the through hole **20***c* is covered with the sealing resin **7**. Unlike the present embodiment, it is possible to have a configuration without the through hole **20***c* and the resin through hole **76**. In the present embodiment, the reverse surface **20***b* of the die pad **20** is covered with the sealing resin **7** as shown in FIGS. **3** to **5**. However, the reverse surface **20***b* may not be covered with the sealing resin **7**, and may be exposed from the resin reverse surface **72** of the sealing resin **7**.

(50) In the present embodiment, the width (the dimension in the third direction x in FIG. **2**) of the third terminal **23** near the base end thereof is larger than the width of each of the first terminal **21**, second terminal **22**, and the fourth terminal **24**. However, the width of the third terminal **23** near the base end thereof may be approximately the same as the width of each of the first terminal **21**, the second terminal **22**, and the fourth terminal **24**. In this case, the widths of the respective portions of the first terminal **21**, the second terminal **22**, the third terminal **23**, and the fourth terminal **24** that are exposed from the sealing resin **7** are uniform.

(51) FIG. **6** is a block diagram showing the circuit configuration of the semiconductor device A**1** of the present embodiment. As described with reference to FIG. **2**, the fourth terminal **24** is electrically connected to the first electrode **151** of the temperature detection diode **15**. The second terminal **22**, which is a source terminal, is electrically connected to the second electrode **152** of the temperature detection diode **15**.

(52) The following describes advantages of the present embodiment.

(53) The semiconductor device A**1** according to the present embodiment includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, and also includes the fourth terminal **24**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the second terminal **22** in the present embodiment) via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the second terminal **22** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present embodiment, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while

the switching element **1** is driven.
(54) In the present embodiment, the second electrode **152** of the temperature detection diode **15** is electrically connected to the second terminal **22**, which is a source terminal. The source terminal (the second terminal **22**) is connected to the ground, which is a reference potential, and the potential is stable at substantially 0 V. The second terminal **22** is also used as the terminal of the temperature detection diode **15**, so that the junction temperature can be measured stably even when a current is supplied to the temperature detection diode **15**. such a configuration is suitable for suppressing an increase in the number of terminals as well as measuring the junction temperature stably when driving the switching element **1**.
(55) Regarding the first terminal **21**, the second terminal **22**, and the third terminal **23** corresponding to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, the first terminal **21** (gate terminal) and the second terminal **22** (source terminal) are at the outermost positions that are opposite to each other in the third direction x. The third terminal **23** (drain terminal) is positioned between the first terminal **21** and the second terminal **22** in the third direction x. According to such a configuration, the arrangement of the first terminal **21**, the second terminal **22**, and the third terminal **23** (the gate terminal, the source terminal, and the drain terminal) is the same as in a conventional three-terminal switching device (semiconductor device). This makes it easy to handle mounting on a circuit board or the like.
(56) The first distance d**13** in the third direction x between the center line C**1** of the first terminal **21** and the center line C**3** of the third terminal **23** is larger than the second distance d**34** in the third direction x between the center line C**3** of the third terminal **23** and the center line C**4** of the fourth terminal **24**, and is larger than the third distance d**24** in the third direction x between the center line C**4** of the fourth terminal **24** and the center line C**2** of the second terminal **22**. As a result, the fourth terminal **24** is positioned in the middle among the three terminals **23**, **24**, and **22** aligned at relatively small intervals. According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**.
(57) Regarding the fourth terminal **24**, the tip side beyond the bent portion **243** is shifted to one side in the first direction z (the side to which the obverse surface **20***a* of the die pad **20** faces). According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**. Furthermore, it is possible to ensure an appropriate creepage distance between the third terminal **23** (drain terminal) and the second terminal **22** (source terminal) by additionally covering, with an insulating resin applied through a potting process, the exposed part of the fourth terminal **24** that extends between the bent portion **243** and the sealing resin **7**. In this case, withstand voltage can be increased for the third terminal **23** (drain terminal) and the second terminal **22** (source terminal).
(58) The fourth terminal **24** may be configured without the bent portion **243**. In this case, the fourth terminal **24** extends straight along the second direction y, as indicated by the imaginary line in FIG. **1**, and the position of the fourth terminal **24** in the first direction z (the position in the vertical direction) is substantially aligned with the positions of the first terminal **21**, the second terminal **22**, and the third terminal **23** in the first direction z. The same applies to the variations described below, and the fourth terminal **24** can also be similarly configured without the bent portion **243**.
(59) FIG. **7** shows a first variation of the semiconductor device A**1** according to the first embodiment. A semiconductor device A**11** of the present variation is different from the semiconductor device A**1** mainly in the configurations of the first terminal **21** and the third terminal **23**, and in the bonding states of the gate wire **52**, the source wire **53**, and the second wire **62**. In FIG. **7**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line. In FIG. **7** and the subsequent drawings, the elements that are identical or similar to those of the semiconductor device A**1** in the above embodiment are designated by the same

reference signs as in the above embodiment, and the descriptions thereof are omitted as appropriate.

(60) In the semiconductor device A**11** shown in FIG. **7**, the first terminal **21** is connected to the die pad **20**, and extends from the die pad **20** along the second direction y. As shown in FIG. **7**, the first terminal **21** is connected to the die pad **20** at the left end in the third direction x in the figure. The first terminal **21** has an intermediate bent portion **213** and a tip portion **212**. The intermediate bent portion **213** is a portion of the first terminal **21** that is bent such that a portion of the first terminal **21** exposed from the sealing resin **7** is shifted upward from the die pad **20** in the first direction z. The first terminal **21** is electrically connected to the drain electrode **131** via the die pad **20** and the bonding member **3**. The first terminal **21** is the drain terminal of the semiconductor device A**11**.

(61) The third terminal **23** is spaced apart from the die pad **20**, and extends along the second direction y. The third terminal **23** has a third pad **231** and a tip portion **232**. The third pad **231** is closest to the die pad **20** in the second direction y. The source wire **53** is bonded to the third pad **231**. The third terminal **23** is electrically connected to the source electrode **133** via the source wire **53**. The third terminal **23** is the source terminal of the semiconductor device A**11**. The second wire **62** is bonded to the third pad **231**. The third terminal **23** is electrically connected to the second electrode **152** via the second wire **62**.

(62) The gate wire **52** is bonded to the second pad **221** of the second terminal **22**. The second terminal **22** is electrically connected to the gate electrode **132** via the gate wire **52**. The second terminal **22** is the gate terminal of the semiconductor device A**11**.

(63) The gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the second pad **221** of the second terminal **22**, and electrically connects the gate electrode **132** of the switching element **1** and the second terminal **22** to each other. The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the third pad **231** of the third terminal **23**, and electrically connects the source electrode **133** of the switching element **1** and the third terminal **23** to each other. The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the fourth pad **241** of the fourth terminal **24**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **24** to each other. The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the third pad **231** of the third terminal **23**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the third terminal **23** to each other.

(64) The semiconductor device A**11** includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the drain terminal, the gate terminal, and the source terminal, and also includes the fourth terminal **24**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the third terminal **23** in the present variation) via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the third terminal **23** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present variation, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.

(65) In the present variation, the second electrode **152** of the temperature detection diode **15** is electrically connected to the third terminal **23**, which is a source terminal. The source terminal (the third terminal **23**) is connected to the ground, which is a reference potential, and the potential is stable at substantially 0 V. The third terminal **23** is also used as the terminal of the temperature

detection diode **15**, so that the junction temperature can be measured stably even when a current is supplied to the temperature detection diode **15**. such a configuration is suitable for suppressing an increase in the number of terminals as well as measuring the junction temperature stably when driving the switching element **1**.
(66) The first distance d**13** in the third direction x between the center line C**1** of the first terminal **21** and the center line C**3** of the third terminal **23** is larger than the second distance d**34** in the third direction x between the center line C**3** of the third terminal **23** and the center line C**4** of the fourth terminal **24**, and is larger than the third distance d**24** in the third direction x between the center line C**4** of the fourth terminal **24** and the center line C**2** of the second terminal **22**. As a result, the fourth terminal **24** is positioned in the middle among the three terminals **23**, **24**, and **22** aligned at relatively small intervals. According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**.
(67) As shown in FIG. **7**, the third terminal **23**, which is a source terminal, is adjacent to the first terminal **21**, which is a drain terminal, in the third direction x as viewed in the first direction z. The first distance d**13** between the first terminal **21** and the third terminal **23** in the third direction x is relatively large, so that the creepage distance between the first terminal **21** (drain terminal) and the third terminal **23** (source terminal) can be appropriately provided. This makes it possible to increase withstand voltage for the first terminal **21** (drain terminal) and the third terminal **23** (source terminal).
(68) FIG. **8** shows a second variation of the semiconductor device A**1** according to the first embodiment. A semiconductor device A**12** of the present variation is different from the semiconductor device A**1** in the bonding states of the gate wire **52**, the source wire **53**, and the second wire **62**. In FIG. **8**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line.
(69) In the semiconductor device A**12** shown in FIG. **8**, the first terminal **21**, the second terminal **22**, the third terminal **23**, and the fourth terminal **24** are configured the same as the semiconductor device A**1** (see FIG. **2**).
(70) The source wire **53** is bonded to the first pad **211** of the first terminal **21**. The first terminal **21** is electrically connected to the source electrode **133** via the source wire **53**. The first terminal **21** is the source terminal of the semiconductor device A**12**. The second wire **62** is bonded to the first pad **211**. The first terminal **21** is electrically connected to the second electrode **152** via the second wire **62**.
(71) The gate wire **52** is bonded to the second pad **221** of the second terminal. The second terminal **22** is electrically connected to the gate electrode **132** via the gate wire **52**. The second terminal **22** is the gate terminal of the semiconductor device A**12**.
(72) The third terminal **23** is electrically connected to the drain electrode **131** via the die pad **20** and the bonding member **3**. The third terminal **23** is the drain terminal of the semiconductor device A**12**.
(73) The gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the second pad **221** of the second terminal **22**, and electrically connects the gate electrode **132** of the switching element **1** and the second terminal **22** to each other. The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the first pad **211** of the first terminal **21**, and electrically connects the source electrode **133** of the switching element **1** and the first terminal **21** to each other. The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the fourth pad **241** of the fourth terminal **24**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **24** to each other. The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the first pad **211** of the first terminal **21**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the first terminal **21** to

each other.
(74) The semiconductor device A**12** includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the source terminal, the gate terminal, and the drain terminal, and also includes the fourth terminal **24**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the first terminal **21** in the present variation) via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the first terminal **21** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present variation, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.
(75) In the present variation, the second electrode **152** of the temperature detection diode **15** is electrically connected to the first terminal **21**, which is a source terminal. The source terminal (the first terminal **21**) is connected to the ground, which is a reference potential, and the potential is stable at substantially 0 V. The first terminal **21** is also used as the terminal of the temperature detection diode **15**, so that the junction temperature can be measured stably even when a current is supplied to the temperature detection diode **15**. such a configuration is suitable for suppressing an increase in the number of terminals as well as measuring the junction temperature stably when driving the switching element **1**.
(76) Regarding the first terminal **21**, the second terminal **22**, and the third terminal **23** corresponding to the three terminals, i.e., the source terminal, the gate terminal, and the drain terminal, the first terminal **21** (source terminal) and the second terminal **22** (gate terminal) are at the outermost positions that are opposite to each other in the third direction x. The third terminal **23** (drain terminal) is positioned between the first terminal **21** and the second terminal **22** in the third direction x. According to such a configuration, the arrangement of the first terminal **21**, the second terminal **22**, and the third terminal **23** (the source terminal, the gate terminal, and the drain terminal) is the same as in a conventional three-terminal switching device (semiconductor device). This makes it easy to handle mounting on a circuit board or the like.
(77) The first distance d**13** in the third direction x between the center line C**1** of the first terminal **21** and the center line C**3** of the third terminal **23** is larger than the second distance d**34** in the third direction x between the center line C**3** of the third terminal **23** and the center line C**4** of the fourth terminal **24**, and is larger than the third distance d**24** in the third direction x between the center line C**4** of the fourth terminal **24** and the center line C**2** of the second terminal **22**. As a result, the fourth terminal **24** is positioned in the middle among the three terminals **23**, **24**, and **22** aligned at relatively small intervals. According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**.
(78) As shown in FIG. **8**, the first terminal **21**, which is a source terminal, is adjacent to the third terminal **23**, which is a drain terminal, in the third direction x as viewed in the first direction z. The first distance d**13** between the first terminal **21** and the third terminal **23** in the third direction x is relatively large, so that the creepage distance between the first terminal **21** (source terminal) and the third terminal **23** (drain terminal) can be appropriately provided. This makes it possible to increase withstand voltage for the first terminal **21** (source terminal) and the third terminal **23** (drain terminal).
(79) FIGS. **9** to **11** show a semiconductor device according to a second embodiment of the present disclosure. A semiconductor device A**2** of the present embodiment includes a substrate **2**A, a first

terminal **41**, a second terminal **42**, a third terminal **43**, and a fourth terminal **44**, instead of the lead frame **2** of the above embodiment. In FIG. **9**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line.

(80) In the present embodiment, the substrate **2**A includes an insulating layer **25**, an obverse-surface conductive layer **26**, and a reverse-surface metal layer **27**. The insulating layer **25** is a plate-like member made of an insulating material. The insulating layer **25** has a rectangular shape as viewed in the first direction z. The material of the substrate **2**A is not particularly limited. For example, the substrate **2**A may be made of a ceramic material such as alumina, aluminum nitride, silicon nitride, boron nitride, or graphite.

(81) The obverse-surface conductive layer **26** forms the upper surface of the substrate **2**A, and is mainly for forming a conductive path to the switching element **1**. The material of the obverse-surface conductive layer **26** is not particularly limited. For example, the obverse-surface conductive layer **26** may be made of a metal such as Cu or an alloy thereof, and may include a plating layer made of Ni or Ag as necessary. The method for forming the obverse-surface conductive layer **26** is not particularly limited. For example, a metal plate member may be bonded to the upper surface of the insulating layer **25**.

(82) As shown in FIGS. **10** and **11**, the obverse-surface conductive layer **26** has an obverse surface **26***a* and a reverse surface **26***b*. The obverse surface **26***a* is the upper surface of the obverse-surface conductive layer **26**. The obverse surface **26***a* (a part of the obverse surface **26***a* corresponding to a drain electrode portion **261** described below) is a surface on which the switching element **1** is mounted. The element reverse surface **12** of the switching element **1** faces the obverse surface **26***a*. The reverse surface **26***b* is the lower surface of the obverse-surface conductive layer **26**. The obverse surface **26***a* and the reverse surface **26***b* are flat and face away from each other in the first direction z.

(83) The obverse-surface conductive layer **26** has a drain electrode portion **261** and a source electrode portion **263**.

(84) The drain electrode portion **261** is a portion on which the switching element **1** is mounted, and to which the third terminal **43** is bonded. The switching element **1** is supported by the insulating layer **25** and the drain electrode portion **261** (obverse-surface conductive layer **26**) via the bonding member **3**. In the present embodiment, the drain electrode portion **261** has a size that occupies most of the obverse-surface conductive layer **26**. The source electrode portion **263** is a portion that is electrically connected to the source electrode **133** of the switching element **1**, and to which the second terminal **42** is bonded. The source electrode portion **263** is spaced apart from the drain electrode portion **261**. The obverse-surface conductive layer **26** configured as described above is an example of a “base”.

(85) As shown in FIGS. **10** and **11**, the reverse-surface metal layer **27** forms the lower surface of the substrate **2**A. The reverse-surface metal layer **27** is insulated from the drain electrode portion **261** and the switching element **1**. In the present embodiment, the reverse-surface metal layer **27** is formed to have the size and shape that cover the most of the lower surface of insulating layer **25**. Hence, the reverse-surface metal layer **27** overlaps with almost the entirety of the drain electrode portion **261** and thus the switching element **1** as viewed in the first direction z.

(86) The first terminal **41**, the second terminal **42**, the third terminal **43**, and the fourth terminal **44** form a conductive path between the outside of the semiconductor device A**2** and the switching element **1**. These terminals **41** to **44** are spaced apart from each other in the third direction x, and are used when the semiconductor device A**2** is mounted on, e.g., a circuit board (not illustrated).

(87) As shown in FIG. **9**, the first terminal **41** extends along the second direction y, and has a bonding portion **411**, a bent portion **412**, and a tip portion **413**. The bonding portion **411** is the root of the first terminal **41**, and is bonded to the upper surface of the substrate **2**A. The method for bonding the bonding portion **411** to the upper surface of the substrate **2**A is not particularly limited. For example, the bonding may be performed with a suitable bonding material selected from among

various bonding materials. The bent portion **412** is bent and connected to the bonding portion **411**, and has a shape that separates the portion between the bent portion **412** and the tip portion **413** apart from the reverse-surface metal layer **27** in the first direction z. The tip portion **413** is the tip of the first terminal **41**, and is located opposite to the bonding portion **411**. The gate wire **52** is bonded to the bonding portion **411**. The first terminal **41** is electrically connected to the gate electrode **132** via the gate wire **52**. In the present embodiment, the first terminal **41** is the gate terminal of the semiconductor device A**2**.

(88) The second terminal **42** extends along the second direction y, and is arranged at the outermost portion (right side in the figure) in the third direction x as viewed in the first direction z. The second terminal **42** has a bonding portion **421**, a bent portion **422**, and a tip portion **423**. The bonding portion **421** is the root of the second terminal **42**, and is bonded to the source electrode portion **263**. The method for bonding the bonding portion **421** and the source electrode portion **263** is not particularly limited, and may be selected appropriately from among various methods such as bonding with a conductive bonding material, ultrasonic bonding, and resistance welding. In the present embodiment, the bonding is performed with a conductive bonding material. The bent portion **422** is bent and connected to the bonding portion **421**, and has a shape that separates the portion between the bent portion **422** and the tip portion **423** apart from the reverse-surface metal layer **27** in the first direction z. The tip portion **423** is the tip of the second terminal **42**, and is located opposite to the bonding portion **421**. The source wire **53** is bonded to the source electrode portion **263**. The second terminal **42** is electrically connected to the source electrode **133** via the source electrode portion **263** and the source wire **53**. In the present embodiment, the second terminal **42** is the source terminal of the semiconductor device A**2**. The second wire **62** is bonded to the source electrode portion **263**. The second terminal **42** is electrically connected to the second electrode **152** via the source electrode portion **263** and the second wire **62**.

(89) The third terminal **43** extends along the second direction y, and is arranged between the first terminal **41** and the second terminal **42** in the third direction x as viewed in the first direction z. The third terminal **43** has a bonding portion **431**, a bent portion **432**, and a tip portion **433**. The bonding portion **431** is the root of the third terminal **43**, and is bonded to the drain electrode portion **261**. The method for bonding the bonding portion **431** and the drain electrode portion **261** is not particularly limited, and may be selected appropriately from among various methods such as bonding with a conductive bonding material, ultrasonic bonding, and resistance welding. In the present embodiment, the bonding is performed with a conductive bonding material. The bent portion **432** is bent and connected to the bonding portion **431**, and has a shape that separates the portion between the bent portion **432** and the tip portion **433** apart from the reverse-surface metal layer **27** in the first direction z. The tip portion **433** is the tip of the third terminal **43**, and is located opposite to the bonding portion **431**. The second terminal **42** is electrically connected to the drain electrode **131** via the drain electrode portion **261** and the bonding member **3**. In the present embodiment, the third terminal **43** is the drain terminal of the semiconductor device A**2**.

(90) The fourth terminal **44** extends along the second direction y, and is arranged between the second terminal **42** and the third terminal **43** in the third direction x as viewed in the first direction z. The fourth terminal **44** has a bonding portion **441**, a bent portion **442**, a tip portion **443**, and a bent portion **444**. The bonding portion **441** is the root of the fourth terminal **44**, and is bonded to the upper surface of the substrate **2**A. The method for bonding the bonding portion **441** to the upper surface of the substrate **2**A is not particularly limited. For example, the bonding may be performed with a suitable bonding material selected from among various bonding materials. The bent portion **442** is bent and connected to the bonding portion **441**, and has a shape that separates the portion between the bent portion **442** and the tip portion **443** apart from the reverse-surface metal layer **27** in the first direction z. The tip portion **443** is the tip of the fourth terminal **44**, and is located opposite to the bonding portion **441**. The bent portion **444** is positioned between the bent portion **442** and the tip portion **443**, and is closer to the bent portion **442**.

(91) The bent portion **444** has a shape that separates the portion between the bent portion **444** and the tip portion **443** apart from the reverse-surface metal layer **27** in the first direction z. As a result, the tip side of the fourth terminal **44**, which is the portion beyond the bent portion **444**, is shifted to one side in the first direction z (the side to which the obverse surface **26***a* of the obverse-surface conductive layer **26** faces). With the fourth terminal **44** having the bent portion **444**, the tip portion **443** of the fourth terminal **44** is shifted to one side in the first direction z (the side to which the obverse surface **26***a* of the obverse-surface conductive layer **26** faces) as compared to the tip portions **413**, **423**, and **433** of the first terminal **41**, the second terminal **42**, and the third terminal **43**.
(92) The first wire **61** is bonded to the bonding portion **441** of the fourth terminal **44**. The fourth terminal **44** is electrically connected to the first electrode **151** via the first wire **61**.
(93) As shown in FIG. **9**, the distance (first distance d**13**) in the third direction x between the center line C**1** of the first terminal **41** (gate terminal) and the center line C**3** of the third terminal **43** (drain terminal) is larger than the distance (second distance d**34**) in the third direction x between the center line C**3** of the third terminal **43** (drain terminal) and the center line C**4** of the fourth terminal **44**. The first distance d**13** is also larger than the distance (third distance d**24**) in the third direction x between the center line C**4** of the fourth terminal **44** and the center line C**2** of the second terminal **42** (source terminal). In the present embodiment, the second distance d**34** and the third distance d**24** are substantially the same. Also, the sum of the second distance d**34** and the third distance d**24** is substantially the same as the first distance d**13**.
(94) As shown in FIG. **9**, the gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the bonding portion **411** of the first terminal **41**, and electrically connects the gate electrode **132** of the switching element **1** and the first terminal **41** to each other. In FIG. **10**, the gate wire **52** is omitted.
(95) The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the source electrode portion **263**, and electrically connects the source electrode **133** of the switching element **1** and the second terminal **42** to each other. In FIGS. **10** and **11**, the source wire **53** is omitted.
(96) The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the bonding portion **441** of the fourth terminal **44**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **44** to each other. In FIG. **11**, the first wire **61** is omitted.
(97) The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the source electrode portion **263** of the second terminal **42**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the second terminal **42**.
(98) The sealing resin **7** covers and protects the switching element **1**, a part of the substrate **2**A, a part of the first terminal **41**, a part of the second terminal **42**, a part of the third terminal **43**, a part of the fourth terminal **44**, the gate wire **52**, the source wire **53**, the first wire **61**, and the second wire **62**.
(99) In the present embodiment, the sealing resin **7** has a resin obverse surface **71**, a resin reverse surface **72**, a pair of resin first side surfaces **73**, and a pair of resin second side surfaces **74**. The resin obverse surface **71** is the upper surface of the sealing resin **7** shown in FIGS. **10** and **11**, and faces in the same direction as the obverse surface **26***a* of the obverse-surface conductive layer **26**. The resin reverse surface **72** is the lower surface of the sealing resin **7** shown in FIGS. **10** and **11**, and faces in the same direction as the reverse surface **26***b* of the obverse-surface conductive layer **26**. The resin obverse surface **71** and the resin reverse surface **72** face away from each other in the first direction z.
(100) As shown in FIGS. **10** and **11**, in the present embodiment, one surface of the reverse-surface metal layer **27** is entirely exposed from the resin reverse surface **72** of the sealing resin **7**. The

surface of the reverse-surface metal layer **27** is flush with the resin reverse surface **72**.

(101) As shown in FIG. **11**, the pair of resin first side surfaces **73** are spaced apart from each other in the second direction y. The pair of resin first side surfaces **73** face away from each other in the second direction y. As shown in FIG. **11**, the upper end of each resin first side surface **73** is connected to the resin obverse surface **71**, and the lower end of each resin first side surface **73** is connected to the resin reverse surface **72**. In the present embodiment, the first terminal **41**, the second terminal **42**, the third terminal **43**, and the fourth terminal partially exposed from one of the resin first side surfaces **73**.

(102) As shown in FIG. **10**, the pair of resin second side surfaces **74** are spaced apart from each other in the third direction x. The pair of resin second side surfaces **74** face away from each other in the third direction x. As shown in FIG. **10**, the upper end of each resin second side surface **74** is connected to the resin obverse surface **71**, and the lower end of each resin second side surface **74** is connected to the resin reverse surface **72**.

(103) The following describes advantages of the present embodiment.

(104) The semiconductor device A**2** according to the present embodiment includes the first terminal **41**, the second terminal **42**, and the third terminal **43**, which correspond to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, and also includes the fourth terminal **44**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **44** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the second terminal **42** in the present embodiment) via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **44** and the second terminal **42** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present variation, the fourth terminal **44**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.

(105) In the present embodiment, the second electrode **152** of the temperature detection diode **15** is electrically connected to the second terminal **42**, which is a source terminal. The source terminal (the second terminal **42**) is connected to the ground, which is a reference potential, and the potential is stable at substantially 0 V. The second terminal **42** is also used as the terminal of the temperature detection diode **15**, so that the junction temperature can be measured stably even when a current is supplied to the temperature detection diode **15**. such a configuration is suitable for suppressing an increase in the number of terminals as well as measuring the junction temperature stably when driving the switching element **1**.

(106) Regarding the first terminal **41**, the second terminal **42**, and the third terminal **43** corresponding to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, the first terminal **41** (gate terminal) and the second terminal **42** (source terminal) are at the outermost positions that are opposite to each other in the third direction x. The third terminal **43** (drain terminal) is positioned between the first terminal **41** and the second terminal **42** in the third direction x. According to such a configuration, the arrangement of the first terminal **41**, the second terminal **42**, and the third terminal **43** (the gate terminal, the source terminal, and the drain terminal) is the same as in a conventional three-terminal switching device (semiconductor device). This makes it easy to handle mounting on a circuit board or the like.

(107) The first distance d**13** in the third direction x between the center line C**1** of the first terminal **41** and the center line C**3** of the third terminal **43** is larger than the second distance d**34** in the third direction x between the center line C**3** of the third terminal **43** and the center line C**4** of the fourth terminal **44**, and is larger than the third distance d**24** in the third direction x between the center line

C**4** of the fourth terminal **44** and the center line C**2** of the second terminal **42**. As a result, the fourth terminal **44** is positioned in the middle among the three terminals **43**, **44**, and **42** aligned at relatively small intervals. According to such a configuration, the fourth terminal **44**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **41** to **43**.

(108) Furthermore, the tip side of the fourth terminal **44**, which is the portion beyond the bent portion **444**, is shifted to one side in the first direction z (the side to which the obverse surface **26***a* of the obverse-surface conductive layer **26** faces). According to such a configuration, the fourth terminal **44**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **41** to **43**. Furthermore, the creepage distance between the third terminal **43** (drain terminal) and the second terminal **42** (source terminal) can be appropriately provided by covering, with an insulating resin through a potting process or the like, a part of the fourth terminal **44** that is exposed from the sealing resin **7** and that extends from the bent portion **444** to the sealing resin **7**. This makes it possible to increase withstand voltage for the third terminal **43** (drain terminal) and the second terminal **42** (source terminal).

(109) The fourth terminal **44** may be configured without the bent portion **444**. In this case, the fourth terminal **44** extends straight along the second direction y, and the position of the fourth terminal **44** in the first direction z (the position in the vertical direction) is substantially aligned with the positions of the first terminal **41**, the second terminal **42**, and the third terminal **43** in the first direction z.

(110) FIGS. **12** and **13** show a semiconductor device according to a third embodiment of the present disclosure. A semiconductor device A**3** of the present embodiment is different from the semiconductor device A**1** in the configuration of the temperature detection diode **15**. In FIG. **12**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line.

(111) In the present embodiment, the first electrode **151** of the temperature detection diode **15** is arranged on the element obverse surface **11**, whereas the second electrode **152** is arranged on the element reverse surface **12**. The pn junction diode portion **150** is formed to extend through the switching element **1** in the thickness direction (first direction z). In the present embodiment, the drain electrode **131** arranged on the element reverse surface **12** also serves as the second electrode **152**. The third terminal **23**, which is a drain terminal, is electrically connected to the drain electrode **131** (second electrode **152**) via the die pad **20** and the bonding member **3**. In the present embodiment, the second wire **62** is not provided.

(112) The semiconductor device A**3** according to the present embodiment includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, and also includes the fourth terminal **24**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the third terminal **23** in the present embodiment). According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the third terminal **23** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present variation, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured temperature detection diode **15** while the switching element **1** is driven.

(113) Regarding the first terminal **21**, the second terminal **22**, and the third terminal **23** corresponding to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, the first terminal **21** (gate terminal) and the second terminal **22** (source terminal) are at

the outermost positions that are opposite to each other in the third direction x. The third terminal **23** (drain terminal) is positioned between the first terminal **21** and the second terminal **22** in the third direction x. According to such a configuration, the arrangement of the first terminal **21**, the second terminal **22**, and the third terminal **23** (the gate terminal, the source terminal, and the drain terminal) is the same as in a conventional three-terminal switching device (semiconductor device). This makes it easy to handle mounting on a circuit board or the like.

(114) The first distance d**13** in the third direction x between the center line C**1** of the first terminal **21** and the center line C**3** of the third terminal **23** is larger than the second distance d**34** in the third direction x between the center line C**3** of the third terminal **23** and the center line C**4** of the fourth terminal **24**, and is larger than the third distance d**24** in the third direction x between the center line C**4** of the fourth terminal **24** and the center line C**2** of the second terminal **22**. As a result, the fourth terminal **24** is positioned in the middle among the three terminals **23**, **24**, and **22** aligned at relatively small intervals. According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**.

(115) Regarding the fourth terminal **24**, the tip side beyond the bent portion **243** is shifted to one side in the first direction z (the side to which the obverse surface **20***a* of the die pad **20** faces). According to such a configuration, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, can be easily distinguished from the other first to third terminals **21** to **23**. Furthermore, the creepage distance between the third terminal **23** (drain terminal) and the second terminal **22** (source terminal) can be appropriately provided by covering, with an insulating resin through a potting process or the like, a part of the fourth terminal **24** that is exposed from the sealing resin **7** and that extends from the bent portion **243** to the sealing resin **7**. This makes it possible to increase withstand voltage for the third terminal **23** (drain terminal) and the second terminal **22** (source terminal).

(116) FIGS. **14** and **15** show a semiconductor device according to a fourth embodiment of the present disclosure. A semiconductor device A**4** of the present embodiment is different from the semiconductor device A**1** in the specific configurations of the first terminal **21** to the fourth terminal **24**, and accordingly, is different from the semiconductor device A**1** in the bonding states of the gate wire **52**, the source wire **53**, the first wire **61**, and the second wire **62**. In FIG. **14**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line.

(117) In the semiconductor device A**4** shown in FIG. **14**, the first terminal **21** is spaced apart from the die pad **20**, and extends to one side (lower side in the figure) in the second direction y. The first terminal **21** is arranged at the outermost position (left side in the figure) in the third direction x as viewed in the first direction z. The first terminal **21** has a first pad **211** and a tip portion **212**. The first terminal **21** is electrically connected to the gate electrode **132** via the gate wire **52**. In the present embodiment, the first terminal **21** is the gate terminal of the semiconductor device A**4**.

(118) The second terminal **22** is spaced apart from the die pad **20**, and is arranged on one side (lower side in FIG. **14**) in the second direction y. The second terminal **22** has a second pad **221** and a plurality of tip portions **222**. The second pad **221** is arranged from the center to the right side in the third direction x as viewed in the first direction z. The plurality of tip portions **222** are arranged at intervals in the third direction x and connected to the second pad **221**. The second terminal **22** (second pad **221**) is electrically connected to the source electrode **133** via the source wire **53**. In the present embodiment, the second terminal **22** is the source terminal of the semiconductor device A**4**. The second wire **62** is bonded to the second pad **221**. The second terminal **22** is electrically connected to the second electrode **152** via the second wire **62**.

(119) The third terminal **23** is connected to the die pad **20** and arranged on the other side (upper side in FIG. **14**) in the second direction y with respect to the die pad **20**. The third terminal **23** is elongated in the third direction x. As shown in FIG. **15**, the third terminal **23** is electrically connected to the drain electrode **131** via the die pad **20** and the bonding member **3**. In the present

embodiment, the third terminal **23** is the drain terminal of the semiconductor device A**4**.
(120) The fourth terminal **24** is spaced apart from the die pad **20**, and extends to one side (lower side in FIG. **14**) in the second direction y. The fourth terminal **24** is arranged closer to the left side in the figure in the third direction x as viewed in the first direction z. The fourth terminal **24** has a fourth pad **241** and a tip portion **242**. The first wire **61** is bonded to the fourth pad **241** of the fourth terminal **24**. The fourth terminal **24** is electrically connected to the first electrode **151** via the first wire **61**.
(121) As shown in FIG. **14**, the gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the first pad **211** of the first terminal **21**, and electrically connects the gate electrode **132** of the switching element **1** and the first terminal **21** to each other. The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the second pad **221** of the second terminal **22**, and electrically connects the source electrode **133** of the switching element **1** and the second terminal **22** to each other. In FIG. **15**, the source wire **53** is omitted.
(122) The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the fourth pad **241** of the fourth terminal **24**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **24** to each other. The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the second pad **221** of the second terminal **22**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the second terminal **22** to each other. In FIG. **15**, the second wire **62** is omitted.
(123) The semiconductor device A**4** includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, and also includes the fourth terminal **24**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to another terminal (the second terminal **22** in the present embodiment) via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the second terminal **22** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present embodiment, the fourth terminal **24**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.
(124) In the present embodiment, the second electrode **152** of the temperature detection diode **15** is electrically connected to the second terminal **22**, which is a source terminal. The source terminal (the second terminal **22**) is connected to the ground, which is a reference potential, and the potential is stable at substantially 0 V. The second terminal **22** is also used as the terminal of the temperature detection diode **15**, so that the junction temperature can be measured stably even when a current is supplied to the temperature detection diode **15**. such a configuration is suitable for suppressing an increase in the number of terminals as well as measuring the junction temperature stably when driving the switching element **1**.
(125) FIG. **16** shows a semiconductor device according to a fifth embodiment of the present disclosure. A semiconductor device A**5** of the present embodiment is different from the semiconductor device A**4** in additionally including a fifth terminal **250**, and accordingly, is different from the semiconductor device A**4** in the bonding state of the second wire **62**. In FIG. **16**, the sealing resin **7** is shown as transparent, and the sealing resin **7** is indicated by an imaginary line.
(126) The fifth terminal **250** is spaced apart from the die pad **20**, and extends to one side (lower side in FIG. **16**) in the second direction y. The fifth terminal **250** is arranged closer to the center in the figure in the third direction x as viewed in the first direction z. The fifth terminal **250** has a fifth

pad **251** and a tip portion **252**. The second wire **62** is bonded to the fifth pad **251**. The fifth terminal **250** is electrically connected to the second electrode **152** via the second wire **62**.
(127) As shown in FIG. **16**, the gate wire **52** is bonded to the gate electrode **132** of the switching element **1** and the first pad **211** of the first terminal **21**, and electrically connects the gate electrode **132** of the switching element **1** and the first terminal **21** to each other. The source wire **53** is bonded to the source electrode **133** of the switching element **1** and the second pad **221** of the second terminal **22**, and electrically connects the source electrode **133** of the switching element **1** and the second terminal **22** to each other.
(128) The first wire **61** is bonded to the first electrode **151** of the switching element **1** (temperature detection diode **15**) and the fourth pad **241** of the fourth terminal **24**, and electrically connects the first electrode **151** of the temperature detection diode **15** and the fourth terminal **24** to each other. The second wire **62** is bonded to the second electrode **152** of the switching element **1** (temperature detection diode **15**) and the fifth pad **251** of the fifth terminal **250**, and electrically connects the second electrode **152** of the temperature detection diode **15** and the fifth terminal **250** to each other.
(129) The semiconductor device A**5** includes the first terminal **21**, the second terminal **22**, and the third terminal **23**, which correspond to the three terminals, i.e., the gate terminal, the source terminal, and the drain terminal, and also includes the fourth terminal **24** and the fifth terminal **250**. The switching element **1** includes the temperature detection diode **15**, and the first electrode **151** of the temperature detection diode **15** is electrically connected to the fourth terminal **24** via the first wire **61**. On the other hand, the second electrode **152** of the temperature detection diode **15** is electrically connected to the fifth terminal **250** via the second wire **62**. According to such a configuration, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the fourth terminal **24** and the fifth terminal **250** that are electrically connected to the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present embodiment, the fourth terminal **24** and the fifth terminal **250**, which are exclusively used for electrical connection with the temperature detection diode **15**, are provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.
(130) The semiconductor device according to the present disclosure is not limited to the foregoing embodiments. Various design changes can be made to the specific structures of the elements in the semiconductor device according to the present disclosure.
(131) For example, FIGS. **17** and **18** show a variation of the semiconductor device according to the third embodiment. In the semiconductor device A**3** according to the third embodiment (see FIGS. **12** and **13**), the second electrode **152** (e.g., the cathode electrode or the anode electrode) of the temperature sensor (temperature detection diode) **15** is made in common with the drain electrode **131** of the switching element **1**. In other words, the second electrode **152** and the drain electrode **131** are directly bonded to each other without the intervention of any individual connecting member such as a wire, or alternatively the second electrode **152** and the drain electrode **131** are integrally formed with each other. In a semiconductor device A**6** according to the present variation, one electrode (cathode **152** in the illustrated example) of the temperature sensor **15** is directly connected to the source electrode **133** of the switching element **1** at the upper surface of the switching element **1**. In other words, in the semiconductor device A**6**, the cathode electrode **152** and the source electrode **133** are made in common within one chip.
(132) Specifically, in the semiconductor device A**6**, a diode **150** at the obverse surface of a SiC substrate (switching element **1**) is used as the temperature sensor **15** (see FIG. **18**). The temperature sensor **15** includes a cathode, which is made of polysilicon doped with p-type impurities, and an anode, which is made of polysilicon doped with n-type impurities. Although an insulating film is formed between the SiC substrate and the temperature sensor **15**, illustration of the insulating film is omitted.

(133) The gate electrode **132** is formed in an active region of the SiC substrate via a gate insulating film. Although the active region includes a source region doped with n-type impurities, and a body region doped with p-type impurities, illustration of these regions is omitted.
(134) An interlayer insulating film is formed on the surface of the SiC substrate, and the source electrode **133**, which is made of a metal such as aluminum, is formed on the interlayer insulating film. The gate electrode **132** is electrically separated from the source electrode **133**, the source region, and the body region by the interlayer insulating film and the gate insulating film. The source electrode **133** is electrically connected to the source region and the body region of the SiC substrate via an opening in the interlayer insulating film.
(135) The anode electrode **151** and the cathode electrode **152** are formed on the upper portion of the temperature sensor **15**. The anode electrode **151** and the cathode electrode **152** are electrically connected to the anode and the cathode of the diode via the opening of the interlayer insulating film, respectively. As described above, in the present variation, the cathode electrode **152** and the source electrode **133** are made in common with each other. In the illustrative example, the cathode electrode **152** and the source electrode **133** are electrically connected to each other via an intermediate conductive connecting part **152***a*. Note that the cathode electrode **152**, the intermediate connecting part **152***a*, and the source electrode **133** are integrally formed as a whole with the use of the same conductive material. In FIGS. **17** and **18**, the dotted lines are used for convenience to indicate the boundaries between these members. Unless otherwise stated, the semiconductor device A**6** has the same configuration as the semiconductor device A**2** according to the second embodiment, which is shown in FIGS. **9** to **11**, for example, except that the two electrodes are made in common. Therefore, description of the same members is omitted here.
(136) According to the configuration shown in FIGS. **17** and **18**, the junction temperature of the switching element **1** can be measured by supplying a current to the temperature detection diode **15** with the use of the terminals **42** and **44** that are electrically connected to the temperature sensor, i.e., the temperature detection diode **15**, measuring the voltage, and using the temperature dependence of the resistance change of the diode. In the present variation, the terminal **44**, which is exclusively used for electrical connection with the temperature detection diode **15**, is provided, whereby the junction temperature can be measured with the temperature detection diode **15** while the switching element **1** is driven.
(137) In the above variation, the source electrode **133** and the source terminal **42** are connected to each other via the wire **53**. However, a metal plate made of copper (e.g., an elongated metal piece having a rectangular cross section) may be used instead of the wire **53**. In this case, the source electrode **133** and the source terminal **42** are bonded to the metal plate via a bonding member such as solder. The connection between the gate electrode **132** and the gate terminal **41** and the connection between the anode electrode **151** and the terminal **44** may be established by the wire **52** and the wire **61** as the illustrated example, or may be established by metal plates instead of the wires **52** and **61**. When the source electrode **133** and the source terminal **42** are bonded to each other via the metal plate, it is preferable that a nickel plating layer or a nickel-palladium-gold plating layer be formed on the source electrode **133** (and thus the cathode electrode **152**). In this case, the same plating structure as the source electrode **133** is formed on each of the anode electrode **151** and the gate electrode **132**.
(138) The present disclosure includes the configurations defined in the following clauses.
(139) Clause 1.
(140) A semiconductor device comprising: a switching element having an element obverse surface and an element reverse surface that face away from each other in a first direction, and including a drain electrode, a gate electrode, and a source electrode, the switching element performing on/off control between the drain electrode and the source electrode by applying a driving voltage across the gate electrode and the source electrode with a potential difference being present between the drain electrode and the source electrode; a base having an obverse surface and a reverse surface

that face away from each other in the first direction, and supporting the switching element with the element reverse surface facing the obverse surface; and a first terminal, a second terminal, a third terminal, and a fourth terminal that each extend in a second direction perpendicular to the first direction, wherein the switching element includes a temperature detection diode having a first electrode provided on the element obverse surface, each of the drain electrode, the gate electrode, and the source electrode is electrically connected to a corresponding one of the first terminal, the second terminal, and the third terminal, and the first electrode is electrically connected to the fourth terminal via a first wire.

Clause 2.

(141) The semiconductor device according to clause 1, wherein the first terminal, the second terminal, the third terminal, and the fourth terminal extend to a first side in the second direction, and are spaced apart from each other in a third direction perpendicular to both of the first direction and the second direction, and the first terminal and the second terminal are positioned at respective outermost positions that are opposite to each other in the third direction, the third terminal is positioned between the first terminal and the second terminal in the third direction, and the fourth terminal is positioned between the second terminal and the third terminal.

Clause 3.

(142) The semiconductor device according to clause 2, wherein the temperature detection diode has a second electrode, and the second electrode is electrically connected to one of the first terminal, the second terminal, and the third terminal.

Clause 4.

(143) The semiconductor device according to clause 3, wherein a first distance in the third direction between the first terminal and the third terminal is larger than a second distance in the third direction between the third terminal and the fourth terminal, and also than a third distance in the third direction between the fourth terminal and the second terminal.

(144) Clause 5.

(145) The semiconductor device according to clause 4, wherein the gate electrode is electrically connected to the first terminal, the source electrode is electrically connected to the second terminal, and the drain electrode is electrically connected to the third terminal.

(146) Clause 6.

(147) The semiconductor device according to clause 5, wherein the second electrode is electrically connected to the second terminal.

(148) Clause 7.

(149) The semiconductor device according to clause 4, wherein the drain electrode is electrically connected to the first terminal, the gate electrode is electrically connected to the second terminal, and the source electrode is electrically connected to the third terminal.

(150) Clause 8.

(151) The semiconductor device according to clause 7, wherein the second electrode is electrically connected to the third terminal.

(152) Clause 9.

(153) The semiconductor device according to clause 3 or 4, wherein the second electrode is arranged on the element obverse surface, and is electrically connected to one of the first terminal, the second terminal, and the third terminal via a second wire.

(154) Clause 10.

(155) The semiconductor device according to clause 3 or 4, wherein the second electrode is arranged on the element reverse surface.

(156) Clause 11.

(157) The semiconductor device according to any of clauses 2 to 4, wherein the gate electrode and the source electrode are arranged on the element obverse surface, and the drain electrode is arranged on the element reverse surface, and each of the gate electrode and the source electrode is

electrically connected to one of the first terminal, the second terminal, and the third terminal via either a gate wire or a source wire.
Clause 12.
(158) The semiconductor device according to any of clauses 2 to 11, further comprising a lead frame including the base and the third terminal, wherein the third terminal extends along the second direction from the base.
(159) Clause 13.
(160) The semiconductor device according to any of clauses 2 to 11, further comprising a substrate, the substrate including an obverse-surface conductive layer forming the base and an insulating layer on which the obverse-surface conductive layer is formed.
(161) Clause 14.
(162) The semiconductor device according to any of clauses 2 to 13, wherein at least a tip portion of the fourth terminal at the first side in the second direction is shifted to a side to which the obverse surface of the base faces in the first direction, as compared to respective tip portions of the first terminal, the second terminal, and the third terminal in the first side in the first direction.
(163) Clause 15.
(164) The semiconductor device according to clause 1, wherein the first terminal, the second terminal, and the fourth terminal are arranged on a first side in the second direction, and are spaced apart from each other in a third direction perpendicular to both of the first direction and the second direction, the third terminal is arranged on a second side in the second direction, and the gate electrode is electrically connected to the first terminal, the source electrode is electrically connected to the second terminal, and the drain electrode is electrically connected to the third terminal.
Clause 16.
(165) The semiconductor device according to clause 15, wherein the temperature detection diode has a second electrode, and the second electrode is electrically connected to the second terminal.
Clause 17.
(166) The semiconductor device according to clause 15, further comprising a fifth terminal extending to the first side in the second direction, wherein the temperature detection diode has a second electrode, and the second electrode is electrically connected to the fifth terminal.
Clause 18.
(167) The semiconductor device according to any of clauses 15 to 17, further comprising a lead frame including the base and the third terminal, wherein the third terminal extends to the second side in the second direction from the base.
(168) Clause 19.
(169) The semiconductor device according to any of clauses 1 to 18, further comprising a sealing resin covering the base, a part of each of the first to fourth terminals, and the switching element.
(170) Clause 20.
(171) The semiconductor device according to any of clauses 1 to 19, wherein the switching element is a SiC switching element.
(172) Clause 21.
(173) A semiconductor device comprising: a switching element; a first external terminal; a second external terminal; a third external terminal; and a fourth external terminal, wherein the switching element includes: an element obverse surface and an element reverse surface that face away from each other in a first direction; a first main electrode (source electrode/emitter electrode) on the element obverse surface; a gate electrode on the element obverse surface; a second main electrode (drain electrode/collector electrode) on the element reverse surface; a temperature detection diode including an anode and a cathode; and a first electrode that is electrically connected to one of the anode and the cathode and arranged on the element obverse surface, the first external terminal is electrically connected to the second main electrode, the second external terminal is electrically connected to the gate electrode, the third external terminal is electrically connected to the first main

electrode, the fourth external terminal is electrically connected to the first electrode, and the anode is electrically connected to one of the first external terminal and the fourth external terminal, and the cathode is electrically connected to the other one of the first external terminal and the fourth external terminal.
Clause 22.
(174) The semiconductor device according to clause 21, further comprising a sealing resin that covers the switching element and a part of each of the first external terminal, the second external terminal, the third external terminal, and the fourth external terminal.
(175) Clause 23.
(176) The semiconductor device according to clause 21 or 22, a first conductive member connecting the further comprising: first main electrode and the third external terminal; a second conductive member connecting the gate electrode and the second external terminal; and a third conductive member connecting the first electrode and the fourth external terminal.
Clause 24.
(177) The semiconductor device according to clause 23, wherein the first conductive member, the second conductive member, and the third conductive member are either bonding wires or metal plates, the bonding wires are made of either aluminum or copper, and the metal plates are made of copper.
(178) Clause 25.
(179) The semiconductor device according to clause 24, further comprising bonding members for fixing the metal plates to bonding targets.
(180) Clause 26.
(181) The semiconductor device according to clause 25, wherein the bonding members are made of solder.
(182) Clause 27.
(183) The semiconductor device according to any of clauses 21 to 26, configured to perform on/off control between the first main electrode and the second main electrode by applying a driving voltage across the gate electrode and the first main electrode with a potential difference between the first main electrode and the second main electrode.
(184) Clause 28.
(185) The semiconductor device according to any of clauses 21 to 27, further comprising a base facing the element reverse surface and supporting the switching element.
(186) Clause 29.
(187) A semiconductor device comprising: a switching element; a first external terminal; a second external terminal; a third external terminal; and a fourth external terminal, wherein the switching element includes: an element obverse surface and an element reverse surface that face away from each other in a first direction; a first main electrode (source electrode/emitter electrode) on the element obverse surface; a gate electrode on the element obverse surface; a second main electrode (drain electrode/collector electrode) on the element reverse surface; a temperature detection diode including an anode and a cathode; and a first electrode that is electrically connected to one of the anode and the cathode and arranged on the element obverse surface, the first external terminal is electrically connected to the second main electrode, the second external terminal is electrically connected to the gate electrode, the third external terminal is electrically connected to the first main electrode, the fourth external terminal is electrically connected to the first electrode, and the anode is electrically connected to one of the third external terminal and the fourth external terminal, and the cathode is electrically connected to the other one of the third external terminal and the fourth external terminal.
Clause 30.
(188) The semiconductor device according to clause 29, further comprising a sealing resin that covers the switching element and a part of each of the first external terminal, the second external

terminal, the third external terminal, and the fourth external terminal.
(189) Clause 31.
(190) The semiconductor device according to clause 29 or 30, further comprising: a first conductive member connecting the first main electrode and the third external terminal; a second conductive member connecting the gate electrode and the second external terminal; and a third conductive member connecting the first electrode and the fourth external terminal.
Clause 32.
(191) The semiconductor device according to clause 31, wherein the first conductive member, the second conductive member, and the third conductive member are either bonding wires or metal plates, the bonding wires are made of either aluminum or copper, and the metal plates are made of copper.
(192) Clause 33.
(193) The semiconductor device according to clause 32, further comprising bonding members for fixing the metal plates to bonding targets.
(194) Clause 34.
(195) The semiconductor device according to clause 33, wherein the bonding members are made of solder.
(196) Clause 35.
(197) The semiconductor device according to any of clauses 29 to 34, configured to perform on/off control between the first main electrode and the second main electrode by applying a driving voltage across the gate electrode and the first main electrode with a potential difference between the first main electrode and the second main electrode.
(198) Clause 36.
(199) The semiconductor device according to any of clauses 29 to 35, further comprising a base facing the element reverse surface and supporting the switching element.
REFERENCE SIGNS
(200) A**1**, A**11**, A**12**, A**2**, A**3**, A**4**, A**5**, A**6**: Semiconductor device **1**: Switching element **11**: Element obverse surface **12**: Element reverse surface **131**: Drain electrode **132**: Gate electrode **133**: Source electrode **15**: Temperature detection diode **150**: pn junction diode portion **151**: First electrode **152**: Second electrode **2**: Lead frame **2**A: Substrate **20**: Die pad (Base) **20***a*: Obverse surface **20***b*: Reverse surface **20***c*: Through hole **21**: First terminal **22**: Second terminal **23**: Third terminal **24**: Fourth terminal **250**: Fifth terminal **211**: First pad **221**: Second pad **231**: Third pad **241**: Fourth pad **251**: Fifth pad **212**, **222**, **232**, **242**, **252**: Tip portion **213**: Intermediate bent portion **233**: Intermediate bent portion **243**: Bent portion **25**: Insulating layer **26**: Obverse-surface conductive layer **26***a*: Obverse surface **26***b*: Reverse surface **261**: Drain electrode portion **263**: Source electrode portion **27**: Reverse-surface metal layer **3**: Bonding member **41**: First terminal **42**: Second terminal **43**: Third terminal **44**: Fourth terminal **411**, **421**, **431**, **441**: Bonding portion **412**, **422**, **432**, **442**: Bent portion **413**, **423**, **433**, **443**: Tip portion **444**: Bent portion **61**: First wire **62**: Second wire **7**: Sealing resin **71**: Resin obverse surface **72**: Resin reverse surface **73**: Resin first side surface **74**: Resin second side surface **75**: Recess **76**: Resin through hole C**1**, C**2**, C**3**, C**4**: Center line d**13**: First distance d**24**: Third distance d**34**: Second distance x: Third direction y: Second direction z: First direction

Claims

1. A semiconductor device comprising: a switching element having an element obverse surface and an element reverse surface that face away from each other in a first direction, and including a drain electrode, a gate electrode, and a source electrode, the switching element performing on/off control between the drain electrode and the source electrode by applying a driving voltage across the gate electrode and the source electrode with a potential difference being present between the drain

electrode and the source electrode; a base having an obverse surface and a reverse surface that face away from each other in the first direction, and supporting the switching element with the element reverse surface facing the obverse surface; and a first terminal, a second terminal, a third terminal, and a fourth terminal that each extend in a second direction perpendicular to the first direction, wherein the switching element includes a temperature detection diode having a first electrode and a second electrode provided on the element obverse surface, the temperature detection diode includes a pn junction diode portion built into the switching element by a semiconductor process, each of the drain electrode, the gate electrode, and the source electrode is electrically connected to a corresponding one of the first terminal, the second terminal, and the third terminal, the first electrode is electrically connected to the fourth terminal, but is connected to none of the first terminal, the second terminal, and the third terminal, and the second electrode is electrically connected to one of the first terminal, the second terminal, and the third terminal.
2. The semiconductor device according to claim 1, wherein the first terminal, the second terminal, the third terminal, and the fourth terminal extend to a first side in the second direction, and are spaced apart from each other in a third direction perpendicular to both of the first direction and the second direction, and the first terminal and the second terminal are positioned at respective outermost positions that are opposite to each other in the third direction, the third terminal is positioned between the first terminal and the second terminal in the third direction, and the fourth terminal is positioned between the second terminal and the third terminal.
3. The semiconductor device according to claim 1, wherein a first distance in the third direction between the first terminal and the third terminal is larger than a second distance in the third direction between the third terminal and the fourth terminal, and also than a third distance in the third direction between the fourth terminal and the second terminal.
4. The semiconductor device according to claim 3, wherein the gate electrode is electrically connected to the first terminal, the source electrode is electrically connected to the second terminal, and the drain electrode is electrically connected to the third terminal.
5. The semiconductor device according to claim 4, wherein the second electrode is electrically connected to the second terminal.
6. The semiconductor device according to claim 3, wherein the drain electrode is electrically connected to the first terminal, the gate electrode is electrically connected to the second terminal, and the source electrode is electrically connected to the third terminal.
7. The semiconductor device according to claim 6, wherein the second electrode is electrically connected to the third terminal.
8. The semiconductor device according to claim 3, wherein the second electrode is arranged on the element obverse surface, and is electrically connected to one of the first terminal, the second terminal, and the third terminal via a second wire.
9. The semiconductor device according to claim 3, wherein the second electrode is arranged on the element reverse surface.
10. The semiconductor device according to claim 2, wherein the gate electrode and the source electrode are arranged on the element obverse surface, and the drain electrode is arranged on the element reverse surface, and each of the gate electrode and the source electrode is electrically connected to one of the first terminal, the second terminal, and the third terminal via either a gate wire or a source wire.
11. The semiconductor device according to claim 2, further comprising a lead frame including the base and the third terminal, wherein the third terminal extends along the second direction from the base.
12. The semiconductor device according to claim 2, further comprising a substrate, the substrate including an obverse-surface conductive layer forming the base and an insulating layer on which the obverse-surface conductive layer is formed.
13. The semiconductor device according to claim 2, wherein at least a tip portion of the fourth

terminal at the first side in the second direction is shifted to a side to which the obverse surface of the base faces in the first direction, as compared to respective tip portions of the first terminal, the second terminal, and the third terminal in the first side in the first direction.
14. The semiconductor device according to claim 1, wherein the first terminal, the second terminal, and the fourth terminal are arranged on a first side in the second direction, and are spaced apart from each other in a third direction perpendicular to both of the first direction and the second direction, the third terminal is arranged on a second side in the second direction, and the gate electrode is electrically connected to the first terminal, the source electrode is electrically connected to the second terminal, and the drain electrode is electrically connected to the third terminal.
15. The semiconductor device according to claim 14, wherein the second electrode is electrically connected to the second terminal.
16. The semiconductor device according to claim 14, further comprising a fifth terminal extending to the first side in the second direction, wherein the second electrode is electrically connected to the fifth terminal.
17. The semiconductor device according to claim 14, further comprising a lead frame including the base and the third terminal, wherein the third terminal extends to the second side in the second direction from the base.
18. The semiconductor device according to claim 1, further comprising a sealing resin covering the base, a part of each of the first to fourth terminals, and the switching element.
19. The semiconductor device according to claim 1, wherein the switching element is a SiC switching element.
20. The semiconductor device according to claim 1, wherein the temperature detection diode includes an anode and a cathode, the first electrode is electrically connected to one of the anode and the cathode and arranged on the element obverse surface, and the second electrode is electrically connected to another one of the anode and the cathode.